

INTERNAL TECHNICAL PROTOCOL: RESTRICTED ACCESS (TIER 1)

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Subject: Sentinel GraphRAG Root Key Infrastructure & Encryption Standards

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1. Overview of the Sentinel Security Plane

The Sentinel platform utilizes a Hybrid GraphRAG architecture where security is enforced at the data-at-rest, data-in-transit, and context-injection layers. This document outlines the management of cryptographic root keys that govern the identity firewall between Neo4j AuraDB and the LLM inference engine.

2. Hardware Security Module (HSM) Integration

All root encryption keys for the Sentinel backend are stored in a FIPS 140-2 Level 3 compliant Hardware Security Module (HSM).

- **Master Key (MK):** The primary 256-bit AES key used to wrap secondary data keys.
- **Access Control:** Access to the HSM requires Dual-Control (M-of-N) authorization, involving at least two Level 1 Administrators (Employee IDs 1*** series).
- **Key Rotation:** Automated rotation occurs every 90 days. Manual rotation is triggered immediately upon any detected Tier 1 credential compromise.

3. Neo4j Graph Security & Cipher Obfuscation

The "Ground Truth" stored in Neo4j is protected via property-level encryption.

- **Deterministic IVs:** Initialization Vectors (IVs) for policy chunks are unique to each document version to prevent pattern matching attacks.
- **Query Sandboxing:** All Cypher queries generated by the Query Copilot are executed in a read-only transaction buffer that is isolated from the main administrative schema.
- **The RBAC Firewall:** Access codes (1 = Confidential, 2 = General) are hard-coded into the node properties and validated against the employee_id prefix at the database kernel level, prior to any data leaving the graph.

4. LLM Context Isolation (Anti-Prompt Injection)

To prevent "jailbreaking" or unauthorized data extraction:

- **Prompt Shielding:** Sentinel utilizes a system-level pre-processor that strips any "ignore previous instructions" commands from user queries.

- **Context Truncation:** If a retrieval result contains a higher sensitivity level than the user's `user_tier`, the system performs a Hard Drop, ensuring the LLM never receives the confidential token stream.

5. Root Key Recovery & Disaster Protocol

In the event of a "Total System Lockout" (e.g., TPM/BitLocker reset error as experienced on Jan 3, 2026):

1. **Initiate the Cold-Start Recovery** using the physical split-keys stored in the Hyderabad Secure Vault.
2. **Re-index all :Policy nodes** using a new salt to invalidate all cached vector embeddings.
3. **Audit all Tier 1 Employee nodes** for unauthorized role escalation.

Authored by:

Sentinel Security Architect

Department of Cyber Governance